

Long-term stability of PEG-based antifouling surfaces in seawater

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Abstract Poly(ethylene glycol) (PEG) is a hydrophilic polymer that has been extensively used in the biomedical and marine environment due to its antifouling properties. In the biomedical field, PEG has been successfully used to functionalize surfaces due to its resistance to cell and nonspecific protein adsorption. However, the long-term stability of PEG has limited its use in some areas. In the shipping industry, there is a great need for long-term solutions to keep the hulls of the ships fouling-free. The long-term stability of PEG in polydimethylsiloxane (PDMS) fouling-release coatings is studied here, in both accelerated laboratory tests and real seawater conditions. This article shows how PEG-based copolymers, which have been exposed in fouling-release coatings to real-life seawater conditions, are isolated and compared to those exposed to accelerated laboratory testing with successful results. The influence of the chemistry of the PEG compounds, the chosen laboratory degrading agents, and the possible degradation pathways and products are discussed.

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Introduction

Biofouling is the accumulation of organisms such as microbes, animals, and plants on immersed surfaces.^{1,2} In biomedical implants, biofouling can cause inflammatory, infectious, and thrombogenic problems.³ Biosensors, membranes, and drug release systems have also been traditionally challenged by the presence of biofouling.³ In the shipping industry, the accumulation of biofouling on ship hulls results in higher frictional resistance, lower maneuverability, and increased dry docking time, with the consequent increase in fuel consumption and other operational costs.¹ It has been reported⁴ that 300 million tons of fuel are consumed yearly (2012) around the world by waterborne transportation and Shultz⁵ studied the impact of biofouling on the US Navy's ships and estimated that heavy fouling can increase the required shaft power of an Oliver Hazard Perry class frigate (FFG-7) with up to 86%.

In order to minimize the amount of fouling on the ship hulls, fouling-release coatings have been used due to the weak adhesion between the fouling organisms and the coating surface.² These coatings have been further functionalized with chemical groups that have proven their resistance toward protein and biofouling adhesion.² For example, Bodkhe et al.^{6,7} have shown that the introduction of amphiphilic and zwitterionic groups on the surface of fouling-release coatings could improve the antifouling performance of the coatings. Likewise, Kavanagh⁸ and Ostuni⁹ obtained similar results when functionalizing surfaces with hydrophilic polymers.

Poly(ethylene glycol) (PEG) has been extensively employed to functionalize surfaces in the biomedical field due to its antifouling properties.¹⁰ Many studies have focused on optimizing the antifouling behavior of

surfaces functionalized with PEG by changing different parameters, mainly (1) the molecular weight (M_w), (2) the chemistry of the end-group, and (3) the grafting density.^{9,11} Over recent years, the widespread use of PEG has also reached the antifouling coatings industry, as supported by the number of recent patents^{12–18} and scientific papers.^{8,19–21} For PEG to be a viable long-term solution in antifouling coatings, it needs to be replenished and retained on the coating surface and stable in the marine environment.

PEG is a linear, hydrophilic, and nonpolar polymer with high resistance to cell and nonspecific protein adsorption.^{10,22} Although not fully understood, the antifouling mechanism of PEG is attributed mainly to (1) its low interfacial energy, (2) hydration due to hydrogen bonding, and (3) the mobility and flexibility of the PEG chains.^{9,22}

In spite of the unique properties of PEG, it also presents some limitations concerning its use in antifouling coatings. Problems related to the long-term stability of PEG have been reported.^{9,23} PEG has shown to undergo both biotic and abiotic degradation under different conditions.^{24,25} In the case of biotic degradation, oxidation of the PEG chain is the most accepted mechanism,²⁴ though others have found evidence of hydrolysis involved in the process.²⁶ The chemical pathway proposed for the biotic degradation of PEG involves the oxidation of the terminal hydroxyl group to an aldehyde and a carboxylic acid, followed by the cleavage of the ether bond, therefore reducing the length of the PEG chain by one unit.²⁴ Very limited research has been done with respect to the biotic degradation of PEG in seawater.²⁷ Regarding the abiotic degradation of PEG, others have proven both photooxidation²⁸ and thermal oxidative degradation^{25,28,29} of PEG in air, both through a radical-based pathway. The suggested mechanism, shown in Fig. 1, is initiated by the formation of a hydroperoxide in a α -C of the oxygen atom in the PEG chain, which is followed by a β -scission, which ends up with the formation of formates and other esters as main products, though formic acid and other products have also been observed.^{25,28–30} Hassouna et al.³¹ and Morlat and Gardette³² compared the results of photooxidation of PEG in air and water. They found similar degradation products, i.e., formates and esters, with the partial hydrolysis of formates into formic acid and formate ions.

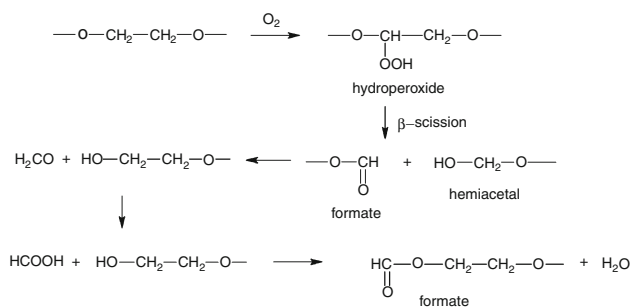


Fig. 1: Chain scission mechanism of PEG under thermal degradation, modified from Mkhathresh & Heatley³⁰ and reproduced with permission from John Wiley and Sons

Nonreactive additives (usually referred to as “oils”) have been successfully added to silicone binders to enhance the antifouling properties. It was first reported by Milne,³³ who added phenyl methyl silicone oil to a silicone fouling-release coating. Others⁸ have followed the same strategy with different polymers. These non-reactive additives migrate to the interface upon exposure, thus changing the surface properties and providing antifouling properties.⁸ PEG has been combined/copolymerized with hydrophobic polymers like polydimethylsiloxane (PDMS) and poly(propylene glycol) (PPG) and used likewise in different coatings. Kavanagh et al.⁸ added PDMS-PEG copolymers as additives to fouling-release silicone coatings, while Wu and Hjort³⁴ employed PEG-PPG-PEG block copolymers in the same way, both with successful results in terms of antifouling properties. Kober and Wesslen²⁰ and Murthy et al.²¹ have also used PEG-based amphiphilic copolymers to prevent protein adsorption on different coatings. Finally, Li et al.³⁵ adsorbed PEG homopolymers and PEG-PPG-PEG copolymers on polystyrene substrates for antifouling purposes, with the block copolymers showing superior properties compared to the PEG homopolymer, probably due to higher stability on the surface.

In spite of the promising use of PEG copolymerized with PPG, it has been reported^{36,37} that PPG is even more prone to radical degradation due to the presence of the methyl side-group. The degradation mechanism proposed for PPG is very similar to PEG (see Fig. 1), and formates, other esters, and alcohols have been reported as main products too.^{36–39} However, there is some discussion whether the secondary^{39,40} or the tertiary³⁷ carbon has the major role in the degradation pathway.

In the present work, copolymers containing PDMS, PEG, and PPG are exposed to (1) different degrading agents in accelerated conditions in the laboratory and (2) real seawater conditions as additives for silicone coatings. The aim of this work is to identify the degradation pathway and products of PDMS-PEG-PPG copolymers under exposure to seawater.

Materials and methods

Materials

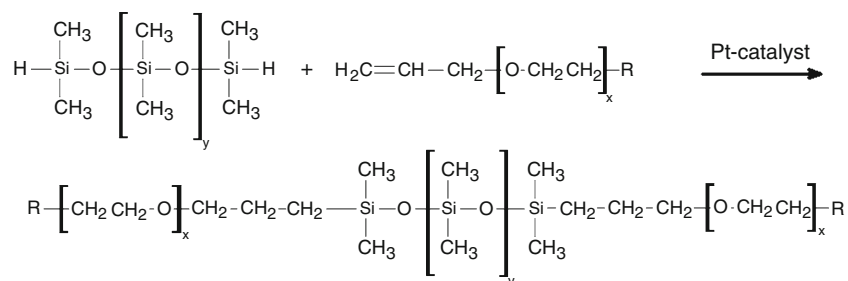
Tetrahydrofuran (THF), heptane (C_7H_{16}), hydrogen peroxide (H_2O_2), hydrochloric acid (HCl), and deuterated chloroform (CCl_3D) were used as received from Sigma-Aldrich. Seawater was collected in Holmen (Copenhagen) and filtered prior to use. Tap water was treated through an osmose water system from Silhorko obtaining slightly acidic distilled water with pH 6.

PDMS-PEG-PPG copolymers

Four different copolymers from different suppliers were used in this study. The structure is triblock X–Y–

Table 1: List of the copolymers studied and their chemical properties

Copolymer	PEG:PPG (weight ratio)	PEG-PPG distribution	End-group
A	2:3	Random	Methoxy
B	1:0	–	Acetate
C	1:0	–	Acetate
D	4:3	Random	Hydroxy

**Fig. 2: Reaction scheme for triblock PEG-PDMS-PEG copolymers, with R being the end-group of the PEG block**

X (X: PEG/PEG-PPG; Y: PDMS) with M_w of 1.000–10.000 g/mol (See Table 1).

Copolymers A, C, and D were used for accelerated testing in the laboratory. Copolymers A and B were used as nonreactive additives in silicone coatings for long-term exposure in seawater.

Separation of nonattached PEG

In the production process of PDMS-PEG-based copolymers, most suppliers make di-hydride-terminated PDMS to react with monoallyl-terminated PEG in the presence of a platinum catalyst, obtaining a copolymer with a propyl linker (a three carbons chain, see (b), (c) and (d) in Fig. 3) between both homopolymers. The reaction scheme is shown in Fig. 2. In that process, the allyl-terminated PEG is usually used in excess, which is not removed when the final product is sold. Heptane was used here prior to the use of the commercial copolymers in order to separate the hydrophilic PEG or PEG-PPG moieties (insoluble in heptane), which were not attached to the PDMS backbone. The process was followed by IR spectroscopy. After laboratory experiments, this separation process was repeated before NMR analysis in order to separate low molecular weight PEG/PPG degradation products. Therefore, two phases were obtained for each experiment: the compounds soluble in heptane (phase 1) and the degradation products of the polyoxyalkylenes (i.e., PEG and PEG-PPG) insoluble in heptane (phase 2).

Laboratory experiments

First, 1 g of each copolymer was exposed to a range of degrading conditions in closed glass containers:

(1) 60 g of HCl solution of pH 2, (2) 60 g of H_2O_2 solution (2 wt%), (3) 60 g of seawater, (4) 60 g of

distilled water, and (5) ultraviolet (UV) radiation. The experiments were performed at 60°C and room temperature (except the experiments with UV radiation that were carried out at 50°C).

Samples from different exposure conditions were obtained and analyzed at different exposure times, from 0 to 105 days for copolymer A and from 0 to 46 days for copolymers C and D.

Seawater experiments

Copolymers A and B were added as additives to conventional silicone (PDMS) coatings. The coatings were prepared by mixing OH-terminated PDMS with $M_w = 40.000$ g/mol with a trifunctional oximinosilane crosslinker (vinyl tris(methyl ethyl ketoxime) silane) in the presence of xylene as solvent. Copolymers A and B were added so they accounted for 3% of the total weight in the dry film. The mixtures were applied on PMMA substrates (10 × 20 cm) using a 8-cm Dr Blake applicator with a 300-μm gap and cured for a week at room temperature. The panels were immersed in static conditions in seawater in Singapore (1 23' 33 N, 103 58' 34 E), due to the high fouling release stress of the area.⁴¹ A sample of each coating was retrieved after 3 and 30 months of exposure.

Liquid chromatography

Semi-preparative Size-Exclusion Chromatography (Semi-prep. SEC) was used to isolate copolymers A and B from PDMS coatings. First, the coatings were immersed in THF for 5 h to extract the nonreacted components from the coating binder (i.e., the studied copolymers and other nonreactive compounds). Then the THF was filtered and run through the Semi-

preparative SEC equipment. The column system consisted of two different columns, Jordi-Gel DVB 500Å (10 × 250 mm) + Jordi-Gel DVB 100Å (10 × 250 mm), followed by a refractive index detector RID-10A, using THF as eluent at a flow rate of 0.8 mL/min at 22°C. An example of the obtained chromatogram is shown in Fig. 8.

NMR

^1H -NMR, ^{13}C -NMR, ^1H -COSY, and ^1H -HSQC were used to analyze the composition of the copolymers, both in laboratory and real exposure experiments. The spectra were recorded on a 300 MHz Bruker spectrometer, using CDCl_3 at 25°C. One ^1H -NMR spectrum of the studied copolymers is shown in Fig. 3 as example.

The loss of PEG groups (relative to the loss of PDMS groups) was calculated as follows:

$$\text{Relative loss of PEG(\%)} = \frac{A_{e,t}/A_{a,t} - A_{e,t0}/A_{a,t0}}{A_{e,t0}/A_{a,t0}} \times 100, \quad (1)$$

where A refers to the area in the ^1H -NMR spectra of hydrogens of every kind (see Fig. 3), the subscript t refers to a certain exposure time, and t_0 refers to the nonexposed samples. Similarly, the relative loss of PPG groups, propyl linkers, and acetate end-groups was calculated. The peak at 2 ppm corresponding to the OCOCH_3 group was used to calculate the relative loss of acetate end-groups for copolymers B and C.

The error bars in Figs. 4a, 4b, and 5 show the standard deviation obtained by three repetitions of a single experiment, assuming that it is representative for all the experiments undertaken.

FTIR spectroscopy

Attenuated Total Reflectance Infrared Spectroscopy (ATR-FTIR) was performed on a ThermoScientific Nicolet iS5 FT-IR. The spectra were recorded with 8 cm^{-1} resolution and 8 scans per sample in the range of $4000\text{--}400\text{ cm}^{-1}$ at room temperature.

UV radiation exposure

The copolymers were exposed to UV radiation in an Accelerated Weathering Tester from QUV (model QUV/se). Lamps UVA-340 were used with an irradiance of $0.89\text{ W/m}^2/\text{nm}$ at 340 nm.

Results and discussion

Laboratory accelerated tests

Copolymers A, C, and D were exposed to accelerated laboratory tests with three main purposes: (1) to screen different degrading agents, (2) to study the effect of the propyl linker and the acetate end-group as possible vulnerable points for radical attack, and (3) to compare

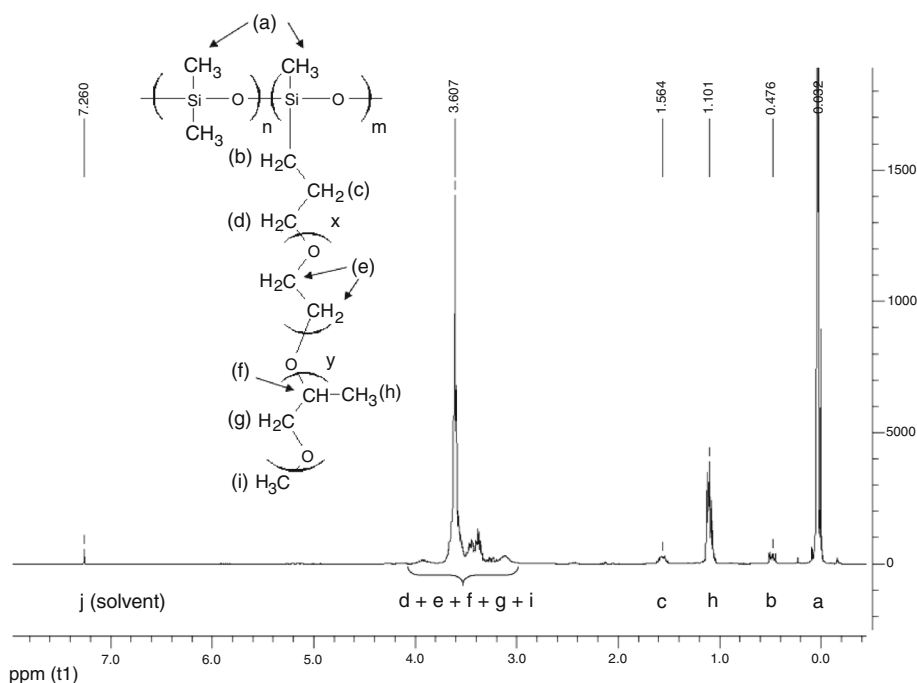


Fig. 3: ^1H -NMR spectrum of copolymer D, with the explanation of different peaks with the corresponding chemical structure

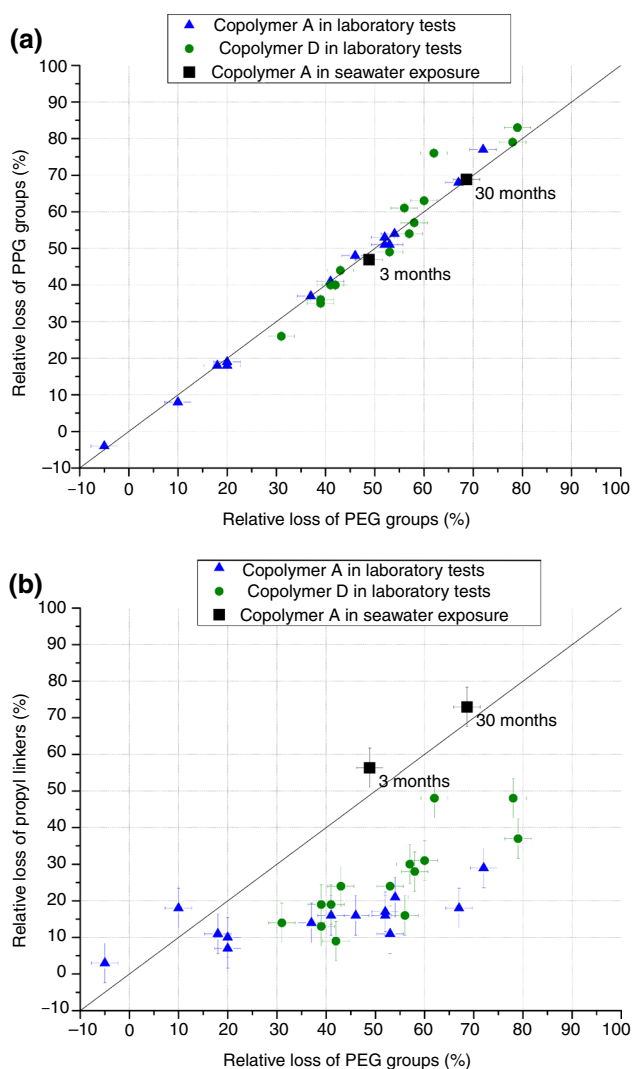


Fig. 4: Relative loss of (a) PPG groups and (b) propyl linkers compared to the loss of PEG groups for copolymers A and D based on $^1\text{H-NMR}$ results. A black line showing an equivalent loss of the two studied groups is added for visual help

the degradation results with copolymers exposed to real-life seawater conditions in silicone coatings.

In Fig. 4, the relative loss of PPG groups and propyl linkers compared to the relative loss of PEG is shown for copolymers A and D for all the degrading conditions used and for different exposure times based on $^1\text{H-NMR}$ results.

Both copolymers A and D show that PEG and PPG units are lost in equal amounts when exposed to the tested degradation conditions, as shown in Fig. 4a. Since the PEG and PPG units are randomly copolymerized, the pattern in Fig. 4a is in agreement with the random chain scission pathway proposed in Fig. 1. There is however another degradation pathway that could explain the behavior seen in Fig. 4a, with the first degradation step taking place in the propyl linker with

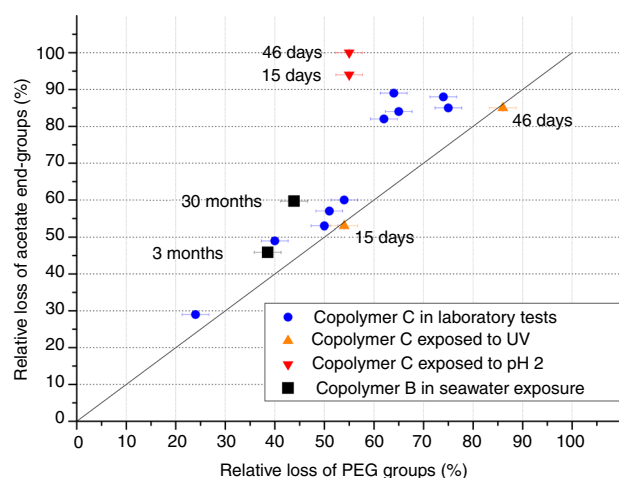


Fig. 5: Relative loss of acetate end-groups compared to the loss of PEG groups for copolymers B and C based on $^1\text{H-NMR}$ results. A black line showing an equivalent loss of the two studied groups is added for visual help

the subsequent scission of the whole polyoxyalkylene chain. That would also lead to the equal loss of PEG and PPG units shown in Fig. 4a.

Figure 4b shows the loss of propyl linkers relative to the loss of PEG units for the same experiments. It can be clearly seen that, though there is a certain loss of propyl linkers, there is no correlation between both variables, thus confirming that the degradation takes place through the proposed random chain scission pathway. No evidence that the propyl linker could be a vulnerable point under the studied degrading conditions is found.

Copolymer D has been used to study the degradation of the acetate end-group of the PEG chain compared to the loss of PEG units. As shown in Fig. 5, the end-group is more prone to degradation as compared to the PEG units. More interestingly, it is possible to observe some differences depending on the degrading agents used. While at pH 2 the degradation occurs on the end-group to a very large extent (95% of the end-groups are lost after 15 days of exposure, while only 55% of the PEG groups are lost), the PEG units and end-groups seem to disappear at the same pace when exposed to UV radiation. No further experiments with a focus on other end-group chemistries were performed in this work.

Some considerations should be made concerning the laboratory test results shown above:

1. Samples exposed to 60°C showed more severe degradation than the corresponding experiments at room temperature. However, no differences were seen in terms of degradation pathways and products. Hence, no differences are observed in Figs. 4 and 5 regarding temperature exposure. Likewise, samples exposed for longer times resulted in higher degradation yields, without further differences, and therefore exposure time is also omitted (unless otherwise stated).

- Copolymers exposed only to seawater or distilled water at 60°C showed degradation of the PEG and PEG/PPG chains. This is in agreement with what has been reported by Morlat and Gardette²⁸ and Gallet et al.³⁸ who successfully thermodegraded PEG and PPG at 50°C in dry conditions. Therefore, the fact of accelerating the degradation tests at 60°C might not be only an acceleration method, but rather a degrading agent by itself.

Analysis of degradation products

Both phase 1 and phase 2 of different experiments (see Materials and Methods) were analyzed by means of IR and NMR in order to identify the degradation products. An example is shown in Fig. 6.

IR and ¹H-NMR results show very little differences in phase 1 (not shown), mainly the loss of PEG and PPG relative to the amount of PDMS. On the contrary, phase 2 shows almost no presence of PDMS and a significant change in composition (see Fig. 6).

Peaks at 4–4.5 ppm (¹H-NMR), 8 ppm (¹H-NMR), and 1725 cm⁻¹ (IR) appear during the degradation process as main degradation peaks in all the studied degradation conditions, though the shapes and sizes of the peaks differ to some extent. ¹³C-NMR was also used to identify the degradation products. An example is shown in Fig. 7 for copolymer C exposed to UV, with the appearance of new peaks around 60–70 and 160–170 ppm. Both NMR and IR results indicate that formates (esters from formic acid) are generated, in agreement with what has been observed by others^{25,29,30}

and as suggested in the mechanism shown in Fig. 1. It has also been observed that the pH of different solutions decreased over time, suggesting the formation of carboxylic acids. Since the carbonyl peak at 1725 cm⁻¹ shows a shoulder and the presence of –OH groups can be observed in the IR spectrum around 3400 cm⁻¹ (see Fig. 6), it cannot be disregarded that a mixture of products consisting of acids, formates, and other esters is present, in agreement with what has been reported by Houssane et al.³¹ The presence of esters as degradation products is also supported by the peak at 174.2 ppm in Fig. 7. Also in agreement with Houssane et al.,³¹ the degradation of the copolymers in aqueous conditions shows less formates and more esters as degradation products compared to the samples exposed in dry conditions, with the IR peak at 1725 cm⁻¹ shifting toward 1735 cm⁻¹. The presence of –OH groups in the IR spectrum, together with the peaks at 61.5 and 72.5 ppm in ¹³C-NMR could also be an indication of the presence of alcohol groups, in agreement with what has been found by Mortensen.⁴³

Seawater experiments

Copolymers A and B were extracted from silicone coatings which had been immersed in seawater as previously explained. After successful isolation of the copolymers using Semi-prep. SEC (see Fig. 8), IR and ¹H-NMR analyses were performed and compared to the unexposed copolymers.

Note that the isolation of the studied copolymer (peak I in Fig. 8) is not complete. This means that

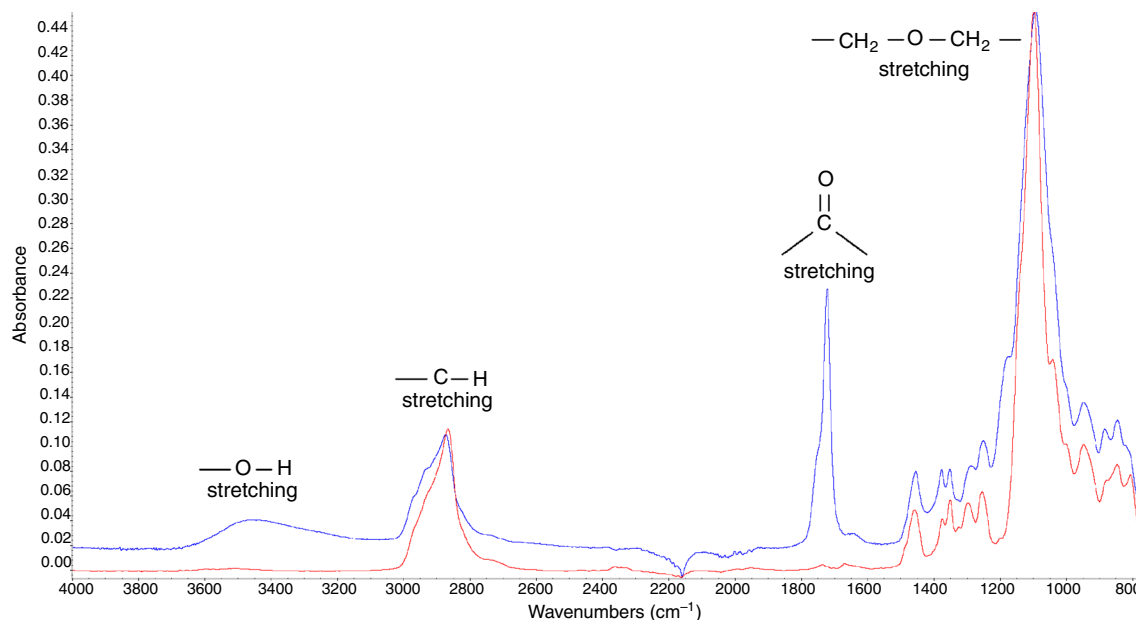


Fig. 6: FTIR spectra of phase 2 of copolymer D before (red) and after (blue) exposure to UV radiation for 8 days at 50°C with the assigned chemistries of the main peaks⁴² (Color figure online)

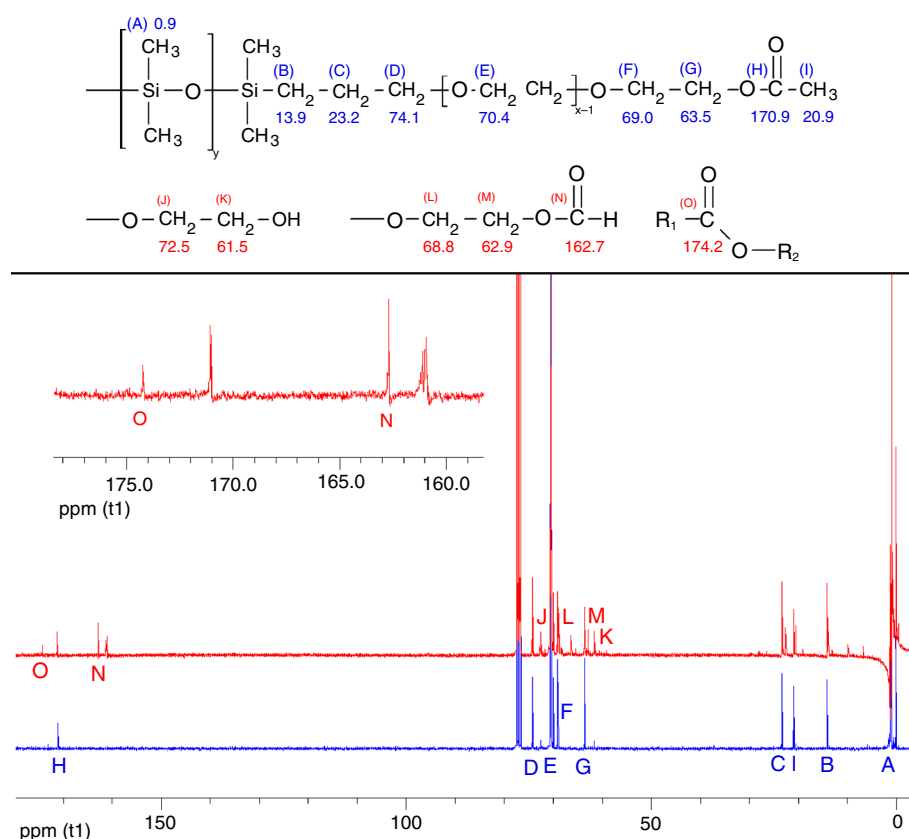


Fig. 7: ^{13}C -NMR spectra of copolymer C unexposed (blue) and exposed to UV radiation for 8 days (red). An amplification of the 160–180 ppm region is shown for the exposed sample. On top, the assignment of different peaks annotated in the spectra of the unexposed (blue) copolymer and the formed degradation products (red). The values of the chemical shifts are expressed in ppm (Color figure online)

some amounts of compounds from peak II might be present when analyzing the isolated copolymers.

The results of the seawater tests are shown as black squares in Figs. 4 and 5, where the losses of PEG, PPG, and propyl linkers for polymer A and PEG and acetate end-groups for copolymer B after 3 and 30 months of exposure are shown.

It should be noted that the apparent decrease in polyoxyalkylene units, propyl linkers, and acetate end-groups found after seawater exposure may arise from an increase in the content of PDMS extracted from the binder system. An increase in PDMS content would lead to an equivalent, but false, decrease of the aforementioned groups, because the loss of all these groups is relative to the amount of PDMS. IR and NMR analyses of peak II of Fig. 8 reveal that PDMS is the main component of this peak. The hypothesis of PDMS extracted from the binder system seems also supported by the fact that most of the degradation takes place in the first three months, with little further degradation occurring in the following 27 months (see Figs. 4 and 5). Ongoing investigations will reveal if

some PDMS extracted from the binder system is altering the results of these experiments.

There is another uncertainty regarding these experiments. If the copolymers studied were degraded, it is reasonable to think that their molecular weight would decrease. Consequently, peak I would contain the copolymer that has not been degraded (or degraded to such a low extent that the change in hydrodynamic volume would be insignificant), while possible degradation products of smaller molecular weight (formates, esters, acids, etc.) would be eluted later within peaks II, III, or IV. Another possibility is that, due to the high solubility of the degradation products of PEG and PPG (i.e., low molecular weight formates, esters, etc.) in water, these products would be released to the seawater environment (instead of staying within the coating film). In order to assess this issue, IR and NMR analyses were performed for all the peaks shown in Fig. 8 to look for any similarities with the degradation products described in the laboratory test results. However, it was not possible to find any sign of those degradation products in neither

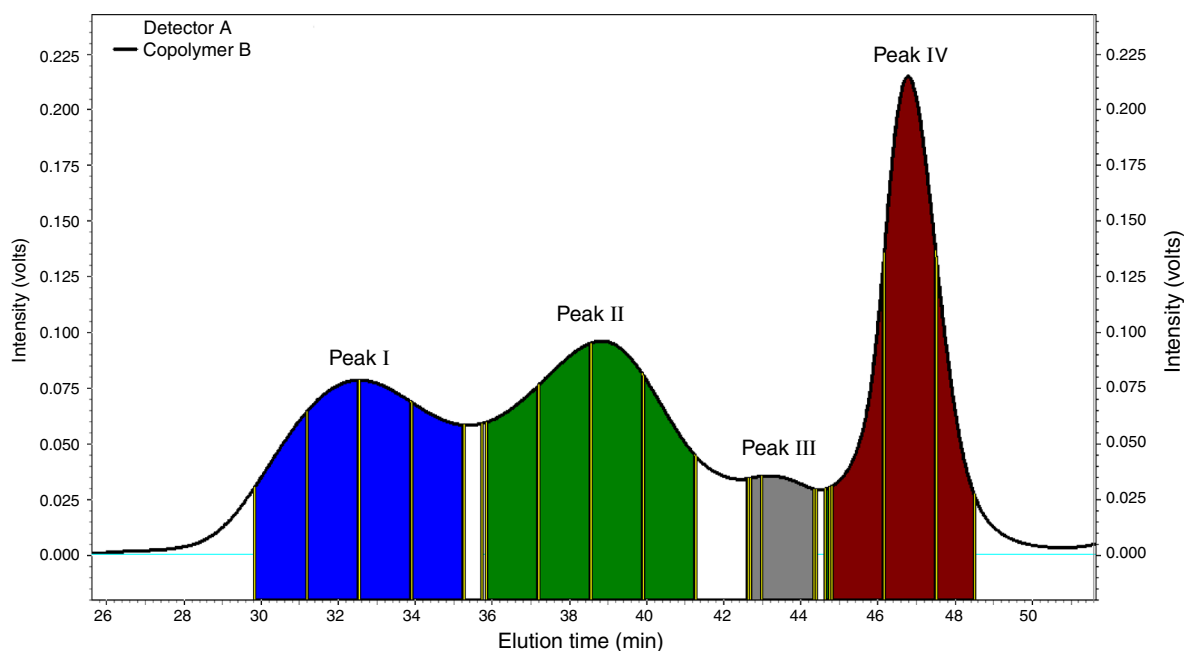


Fig. 8: Size-exclusion chromatogram for the isolation of copolymer B after 30 months of exposure. FTIR and NMR analyses revealed the composition of different peaks: copolymer B (peak I), nonreactive PDMS from the binder system (peak II), and low M_w nonreactive PDMS chains and residues from the curing system (peak III and peak IV)

IR or ^1H -NMR. Whether no degradation has taken place after 30 months of exposure or the degradation products have been eluted into seawater cannot be clarified at this stage.

Conclusions

The degradation of PEG-based copolymers for antifouling purposes has been studied, both in accelerated laboratory conditions and in real seawater conditions. Formates and other esters have been identified as the main products from the laboratory tests, in agreement with other literature studies, though indications of other products have been also found. The relative amounts of each degradation product depend on the degradation agent used, as supported by small differences in NMR and IR results, with formates being the main product in dry conditions while other esters and acids were found in aqueous tests. Different degradation patterns have been identified for the degradation of PEG copolymers with acetate end-group. While the acetate end-groups were severely degraded at pH 2 compared to PEG units, UV radiation degraded acetate and PEG units at the same pace.

PEG-based copolymers (A and B) have been exposed to seawater as additives for silicone coatings. After long-term exposure of these coatings, semi-preparative SEC has been successfully used to isolate the exposed copolymers. The degradation patterns of these copolymers seem to agree with the results obtained for the accelerated laboratory conditions in

terms of loss of PEG, PPG, and acetate end-groups. However, unusually high amounts of degradation in the first 3 months of exposure, together with differences in the degradation behaviors of the propyl linkers, leave some uncertainties regarding the isolation process. It was not possible to analyze the degradation products (if any), which were possibly eluted into seawater along the 30 months of exposure. Furthermore, the influence of the degradation of these copolymers on the antifouling performance of the coatings should be assessed in future experiments.

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